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ELECTRONIC PRINCIPLES AND CIRCUITS **(BEC303)**

MODULE – 2

MOSFET: Biasing | Small Signal Models | Amplifier Configurations

Fixing V_{GS} | Fixing V_G | Drain-to-Gate Feedback | DC Bias Point | Signal Current
Transconductance | T-Model | pi-Model | CS | CG | Source Follower

Department of ECE | VTU B.E. 3rd Semester

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PART A: BIASING IN MOS AMPLIFIER CIRCUITS

1. Biasing by Fixing VGS

1.1 Why Fixing VGS is NOT Recommended

The most straightforward approach to biasing a MOSFET is to fix its gate-to-source voltage **VGS** to the exact value required to provide the desired drain current **ID**. The fixed **VGS** can be derived from the power-supply **VDD** through a voltage divider, or from any other stable reference voltage.

However, this is **NOT a good biasing approach** for the following critical reasons:

- The threshold voltage V_t , the oxide capacitance C_{ox} , and (to a lesser extent) the transistor aspect ratio W/L vary widely among devices of supposedly the same type and size.
- The spread in device parameters is especially large in integrated circuits — particularly among devices fabricated on different wafers or between different production batches.
- Both V_t and the carrier mobility μ_n depend on temperature. If V_{GS} is fixed, the drain current I_D becomes very sensitive to temperature, causing the Q-point to drift.
- For a fixed V_{GS} , two different MOSFETs of the same type can have widely different drain currents I_{D1} and I_{D2} due to process variation — making the bias unreliable.

and note that the values of the threshold voltage V_t , the oxide-capacitance C_{ox} , and (to a lesser extent) the transistor aspect ratio vary widely among devices of supposedly the same size and type. The spread is also large in integrated circuits, especially among devices fabricated on different wafers and certainly between different batches of wafers. Furthermore, both V_t and μ_n depend on temperature, with the result that if we fix the value of V_{GS} , the drain current I_D becomes very much temperature dependent. As shown in the Fig. 5.51 two I_D - V_{DS} characteristic curves representing extreme values in a batch of MOSFETs of the same type having different current I_D for the fixed value of V_{GS} .

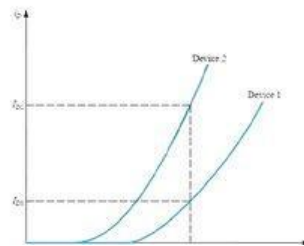


Figure 5.51: Fixed V_{GS} biasing causes large variability in I_D . Device 1 and 2 represent extremes of the same MOSFET type.

1.2 I_D Equation and V_{GS} Dependence

The MOSFET drain current in saturation is given by:

$$I_D = \left(\frac{1}{2}\right) \times k_n \times \left(\frac{W}{L}\right) \times (V_{GS} - V_t)^2 = \left(\frac{1}{2}\right) \times k_n \times V_{OV}^2$$

where $k_n = \mu_n \times C_{ox}$ is the process transconductance parameter, W/L is the aspect ratio, V_t is the threshold voltage, and $V_{OV} = V_{GS} - V_t$ is the overdrive voltage.

Since k_n , V_t , and W/L all have process and temperature variations, fixing V_{GS} results in wide variability of I_D — making this biasing approach unsuitable for production circuits.

2. Biasing by Fixing VG (with Source Resistance RS)

2.1 Circuit Description and VG Equation

An excellent and widely used biasing technique for discrete MOSFET circuits is to fix the dc voltage at the gate **VG** and connect a resistance **RS** in the source lead. This is shown in Figure 5.52(a). For this circuit:

$$V_G = V_{GS} + R_S \times I_D \quad (\text{fundamental } V_G \text{ bias equation})$$

Since **VG** is fixed (set by the voltage divider **RG1** and **RG2**), any change in **ID** directly affects **VGS** through the **RS** term. This provides automatic stabilization of the bias current.

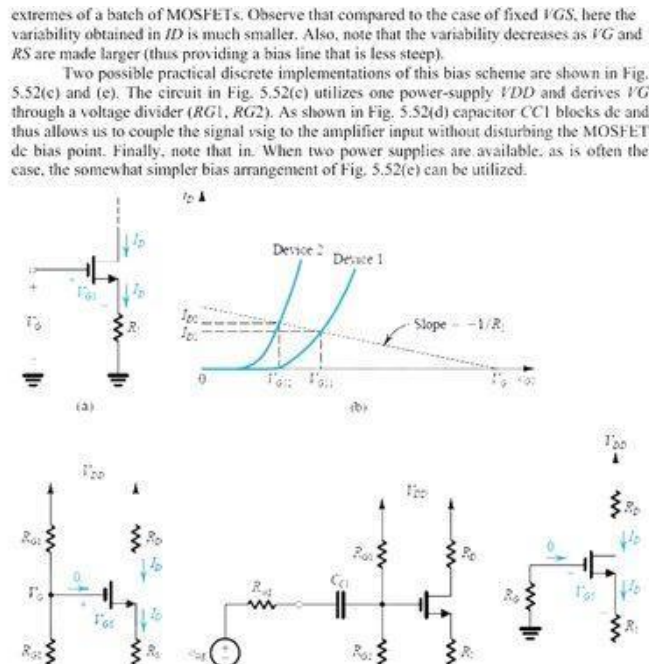


Figure 5.52: Biasing with fixed V_G and source resistance R_S . (a) Basic concept (b) Reduced I_D variability (c) Single-supply implementation with R_{G1} , R_{G2} (d) With coupling cap (e) Dual-supply implementation

2.2 Negative Feedback Mechanism of RS

The source resistor **RS** provides a powerful negative feedback that stabilizes the bias point. Here is how it works:

1. Suppose I_D increases for any reason (temperature rise or transistor with higher g_m)
2. The voltage drop across R_S increases: $V_S = I_D \times R_S$ increases
3. Since V_G is fixed: $V_{GS} = V_G - V_S = V_G - I_D \times R_S$ must decrease
4. Decreased V_{GS} means decreased I_D (from the MOSFET square-law characteristic)
5. This decrease in I_D opposes the original increase — restoring I_D toward its nominal value

This negative feedback action of **RS** is called **source degeneration**. The name "degeneration" comes from the fact that **RS** reduces the effective transconductance of the circuit (degenerates the gain), but in exchange provides valuable bias stability.

NOTE: The variability of I_D decreases as V_G and R_S are made larger — the bias line (slope = $-1/R_S$) becomes less steep, causing the two device curves to intersect at more similar I_D values.

2.3 Practical Implementations

Two practical implementations exist:

- Single power supply V_{DD} : V_G is derived through a voltage divider (R_{G1} , R_{G2}). A coupling capacitor C_{C1} blocks DC and allows AC signal coupling to the gate without disturbing the DC bias.

- Dual power supply (VDD and $-V_{SS}$): A simpler bias arrangement is possible. The gate is connected directly through R_G to ground. The negative supply at the source provides the required gate-source voltage through R_S .

3. Biasing Using Drain-to-Gate Feedback Resistor

3.1 Circuit Description and Operation

A simple and effective biasing arrangement uses a large feedback resistor R_G connected directly between the drain and gate terminals, as shown in Figure 5.54. Since the gate current is zero ($I_G = 0$), there is no voltage drop across R_G , which forces the gate voltage to equal the drain voltage:

$$V_{GS} = V_{DS} = V_{DD} - I_D R_D$$

This can be rewritten as:

$$V_{DD} = V_{GS} + R_D I_D \quad (\text{same form as the fixing-VG equation})$$

which can be rewritten in the form

$$V_{DD} = V_{GS} + R_D I_D$$

which is identical in form to Eq. (5.93), which describes the operation discussed above [that in Fig. 5.52(a)]. Thus, here too, if I_D for some reason increases, then Eq. (5.96) indicates that V_{GS} must decrease. The decrease in V_{GS} causes a decrease in I_D , a change that is opposite in direction to the original increase. Thus the negative feedback or degeneration provided by R_G works as constant as possible.

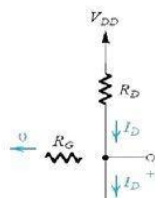


Figure 5.54: Drain-to-gate feedback biasing using large resistance R_G .

3.2 Feedback Mechanism and Stability

The drain-to-gate resistor provides self-stabilizing negative feedback. The mechanism is:

6. Suppose I_D increases for any reason
7. The drain voltage $V_D = V_{DD} - I_D R_D$ decreases
8. Since $V_G = V_D$ (connected by R_G), V_G also decreases
9. This means $V_{GS} = V_G - V_S = V_G$ (since $V_S = 0$) decreases
10. Decreased V_{GS} decreases I_D — opposing the original increase

The feedback provided by R_G is equivalent in form to the source degeneration in the fixing-VG method. Both provide self-correcting action that keeps I_D stable despite transistor variations.

KEY: Drain-to-gate bias forces $V_{GS} = V_{DS}$, which ensures the MOSFET stays in saturation (since $V_{DS} = V_{GS} > V_{GS} - V_t = V_{OV} > 0$ automatically), making it a very simple and self-consistent bias scheme.

PART B: SMALL SIGNAL OPERATION AND MODELLING

4. The DC Bias Point

For small-signal operation, we use the conceptual MOSFET amplifier circuit (Figure 5.34). A dc voltage V_{GS} biases the transistor at its operating point, and the AC input signal v_{gs} is superimposed on this dc bias. The output is taken at the drain terminal.

Figure 5.54 Biasing the MOSFET using a large drain-to-gate feedback

Small-Signal Operation and Models

For small signal operation we utilize the conceptual amplifier circuit 5.34. Here the MOS transistor is biased by applying a dc voltage V_{GS} , and the amplified, v_{gs} , is superimposed on the dc bias voltage V_{GS} . The output v_o is taken at the drain.

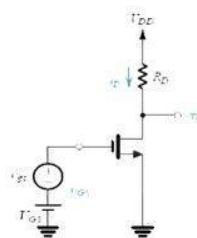


Figure 5.34 Conceptual circuit utilized to study the MOSFET as a small-signal amplifier.

Figure 5.34: Conceptual circuit for MOSFET small-signal amplifier analysis. DC bias V_{GS} sets Q-point; AC signal v_{gs} is superimposed.

The DC bias current I_D is found by setting $v_{gs} = 0$ (no AC signal):

$$I_D = (1/2) \times k_n \times (V_{GS} - V_t)^2 = (1/2) \times k_n \times V_{OV}^2$$

where $V_{OV} = V_{GS} - V_t$ is the overdrive voltage at the Q-point.

The dc voltage at the drain is:

$$V_{DS} = V_{DD} - R_D \times I_D$$

To ensure the MOSFET operates in the saturation region (required for amplification):

$$V_{DS} > V_{OV} \quad (\text{saturation condition: } V_{DS} > V_{GS} - V_t)$$

5. The Signal Current in the Drain Terminal

5.1 Total Drain Current and Small-Signal Condition

When the AC signal v_{gs} is applied, the total gate-to-source voltage becomes:

$$v_{GS} = V_{GS} + v_{gs}$$

Substituting into the MOSFET drain current equation:

$$\begin{aligned} i_D &= (1/2) \times k_n \times (V_{GS} + v_{gs} - V_t)^2 \\ &= (1/2) \times k_n \times (V_{GS} - V_t)^2 + k_n \times (V_{GS} - V_t) \times v_{gs} + (1/2) \times k_n \times v_{gs}^2 \end{aligned}$$

The three terms represent:

- Term 1: $(1/2) \times k_n \times (V_{GS} - V_t)^2 = I_D$ — the DC bias current (quiescent current)
- Term 2: $k_n \times (V_{GS} - V_t) \times v_{gs} = i_d$ — the desired signal current, proportional to v_{gs}
- Term 3: $(1/2) \times k_n \times v_{gs}^2$ — undesired component proportional to $v_{gs}^2 \rightarrow$ causes NONLINEAR DISTORTION

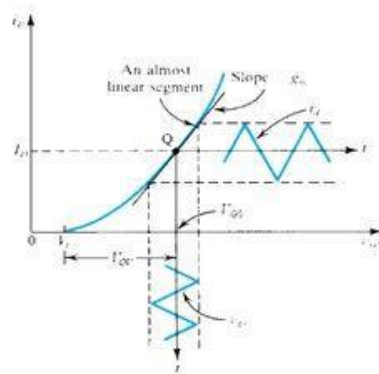


Figure 5.35: Small signal operation of the MOSFET amplifier

The Voltage Gain

Distortion in this circuit of Fig. 5.34 is due to the third (undesired) term in the equation.

Figure 5.35: Small-signal operation of the MOSFET amplifier. Slope of i_D - v_{GS} at Q-point = g_m .

5.2 Nonlinear Distortion and How to Reduce It

The third term $(1/2) \times k_n \times v_{gs}^2$ is proportional to the SQUARE of the input signal — it represents nonlinear (harmonic) distortion that distorts the output waveform. To reduce this distortion:

$$(1/2) \times k_n \times v_{gs}^2 \ll k_n \times (V_{GS} - V_t) \times v_{gs}$$

Simplifying, the **small-signal condition** is:

$$v_{gs} \ll 2(V_{GS} - V_t) = 2 \times V_{OV}$$

When this condition is met, the squared term is negligible, and the total drain current simplifies to:

$$i_D \approx I_D + i_d \quad \text{where } i_d = k_n \times (V_{GS} - V_t) \times v_{gs} = g_m \times v_{gs}$$

6. The Voltage Gain

From the small-signal equivalent circuit, the total drain voltage v_{DS} is:

$$v_{DS} = V_{DD} - R_D \times i_D = V_{DD} - R_D \times (I_D + i_d) = v_{DS} - R_D \times i_d$$

The AC (signal) component of the drain voltage, called v_{ds} , is:

$$v_{ds} = -i_d \times R_D = -g_m \times v_{gs} \times R_D$$

Therefore the **voltage gain** of the MOSFET amplifier is:

$$A_v = v_{ds} / v_{gs} = -g_m \times R_D$$

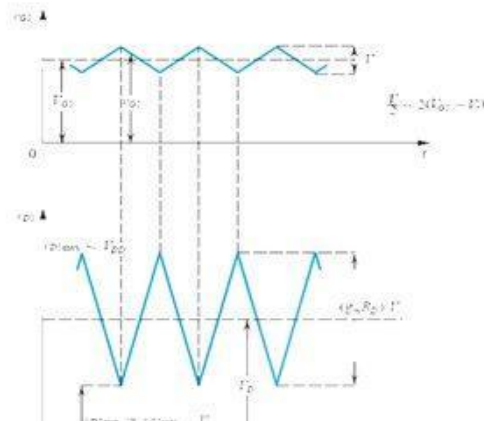


Figure 5.36: Input v_{GS} and output v_{DS} waveforms. Note 180° phase inversion (negative sign in A_v). $v_{GS,max}/2 \ll 2(V_{GS} - V_t)$ for linear operation.

Key Observations:

- The **NEGATIVE** sign indicates 180° phase inversion between input v_{gs} and output v_{ds} — same as CE (BJT) amplifier.
- The magnitude of gain = $g_m \times R_D$. To increase gain: increase g_m (increase I_D or W/L) or increase R_D .
- For linear (undistorted) operation: $v_{DS,min}$ must stay above $v_{GS,max} - V_t$ (remain in saturation) AND $v_{DS,max}$ must stay below V_{DD} (avoid cutoff).

7. Small-Signal Equivalent Circuit Models

7.1 The pi (Hybrid-pi) Model

From a signal perspective, the MOSFET behaves as a **voltage-controlled current source (VCCS)**: it accepts a small AC signal v_{gs} between gate and source, and provides a drain current of $g_m v_{gs}$ at the drain terminal.

The ideal small-signal model has:

- Infinite input resistance at the gate — MOSFET gate draws NO current ($I_G = 0$)
- Controlled current source $g_m v_{gs}$ at the drain
- Infinite output resistance (ideal model, ignoring channel-length modulation)

Figure 5.36 Total instantaneous voltages v_{gs} and v_{ds} for the circuit in Fig. 5.31.

Small-Signal Equivalent-Circuit Models

From a signal point of view, the FET behaves as a voltage-controlled current source. It accepts a signal v_{gs} between gate and source and provides a current $g_m v_{gs}$ at the drain terminal. The input resistance of this controlled source is very high—ideally, infinite. The output resistance that is, the resistance looking into the drain also is high, and we have assumed it to be infinite thus far. Putting all of this together, we arrive at the circuit in Fig. 5.37(a), which represents the small-signal operation of the MOSFET and is thus a **small-signal model** or a **small-signal equivalent circuit**.

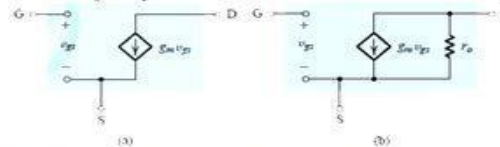


Figure 5.37 Small-signal models for the MOSFET: (a) neglecting the dependence of I_D on v_{DS} in Sat; (b) including it.

Figure 5.37: MOSFET small-signal models. (a) Ideal model neglecting r_o . (b) Complete model including $r_o = |V_A|/I_D$ for channel-length modulation.

7.2 Output Resistance r_o

The ideal model assumes drain current in saturation is completely independent of v_{DS} . In reality, the drain current increases slightly with v_{DS} due to **channel-length modulation** — the effective channel length decreases as v_{DS} increases.

This effect is modeled by a finite resistance r_o (output resistance) connected between drain and source:

$$r_o = |V_A| / I_D$$

where V_A is the **Early voltage** of the MOSFET (analogous to V_A in BJT), typically 10–50V. Including r_o gives the **complete small-signal pi model** shown in Figure 5.37(b).

NOTE: In discrete MOSFET amplifier analysis, r_o is often neglected (assumed infinite) since $R_D \ll r_o$. In IC design, r_o must always be included as it significantly affects performance.

8. The Transconductance g_m

8.1 Three Useful Expressions for g_m

The MOSFET transconductance g_m is the parameter that relates the signal drain current to the signal gate-source voltage. It is defined as:

$$g_m = i_d / v_{gs} \text{ (= slope of } i_D\text{-}v_{GS} \text{ curve at Q-point)}$$

Three equivalent expressions for g_m :

Expression 1 — In terms of V_{GS} and V_t (overdrive voltage):

$$g_m = k_n \times (V_{GS} - V_t) = k_n' \times (W/L) \times V_{OV}$$

Expression 2 — In terms of DC bias current I_D :

$$g_m = 2I_D / (V_{GS} - V_t) = 2I_D / V_{OV}$$

Expression 3 — Combined expression showing square-root dependence:

$$g_m = \sqrt{2 \times k_n' \times (W/L) \times I_D}$$

$$g_m = \sqrt{2k_n' \sqrt{W/L} \cdot I_{DQ}} \quad (5.56)$$

This expression shows two things:

1. For a given MOSFET, g_m is proportional to the square root of the dc bias current.
2. At a given bias current, g_m is proportional to $\sqrt{W/L}$.

Yet another useful expression for g_m of the MOSFET can be obtained by substituting for $k_n'(W/L)$ in Eq. (5.55) by $2I_{DQ}/(V_{GS} - V_t)^2$:

$$g_m = \frac{2I_{DQ}}{V_{GS} - V_t} = \frac{2I_{DQ}}{V_{OV}} \quad (5.57)$$

A convenient graphical construction that clearly illustrates this relationship is shown in Fig. 5.38.

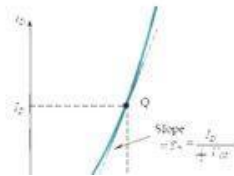


Figure 5.38: g_m is the slope of i_D - v_{GS} at Q-point. The tangent intersects v_{OV} axis at $V_{OV}/2$.

Key insights from these expressions:

- For a given MOSFET: g_m is proportional to the SQUARE ROOT of the dc bias current I_D .
- At a given bias current: g_m is proportional to $\sqrt{W/L}$ — wider, shorter transistors have higher g_m .
- To achieve large g_m : bias at high I_D , use large W/L (wide and short channel transistor).
- Unlike BJT where $g_m = I_C/V_T$ is independent of device geometry, MOSFET g_m depends on both bias current AND device dimensions.

9. The T Equivalent-Circuit Model

Through a simple circuit transformation, an alternative equivalent circuit model called the **T model** can be derived for the MOSFET. The T model is equivalent to the pi model but has a different circuit topology that is more convenient for analyzing CS amplifiers with source resistance and CG amplifiers.

Development of the T Model (Figure 5.40):

11. Start with the standard pi model (Fig. 5.40a): current source $g_m \times v_{gs}$ between drain and source, v_{gs} across gate-source.
12. Add a second identical current source $g_m \times v_{gs}$ in series with the original, flowing in the opposite direction (Fig. 5.40b). This does not change terminal currents.
13. The newly created node X is joined to gate terminal G (Fig. 5.40c). Gate current remains zero — no change.
14. The controlled source $g_m \times v_{gs}$ now has its control voltage v_{gs} across it. By the source-absorption theorem, this can be replaced by a resistance of value $v_{gs}/(g_m \times v_{gs}) = 1/g_m$ (Fig. 5.40d).

above without r_o . In Fig. 5.40(b) we have added a second $g_m v_{gs}$ current source in series with the original controlled source. This addition obviously does not change the terminal currents and is thus allowed. The newly created circuit node, labeled X, is joined to the gate terminal G in Fig. 5.40(c). Observe that the gate current does not change — that is, it remains equal to zero — and thus this connection does not alter the terminal characteristics. We now note that we have a controlled current source $g_m v_{gs}$ connected across its control voltage v_{gs} . We can replace this controlled source by a resistance as long as this resistance draws an equal current as the source. (See the source-absorption theorem in Appendix D.) Thus the value of the resistance is $v_{gs}/g_m v_{gs} = 1/g_m$. This replacement is shown in Fig. 5.40(d), which depicts the alternative model. Observe that i_d is still zero, $i_d = g_m v_{gs}$, and $i_s = -i_d/(1/g_m) = g_m v_{gs}$, all the same as in the original model in Fig. 5.40(a).

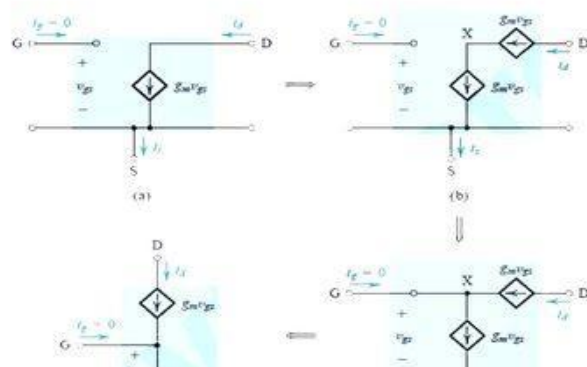


Figure 5.40: Development of T equivalent-circuit model. (a) Standard pi model. (b) Adding series source. (c) Connecting node X to gate. (d) Final T model with $1/g_m$ resistance.

In the T model:

- The resistance $1/g_m$ appears between gate and source (in the source lead)
- The current source $g_m \times v_{gs}$ ($= i_d$) appears at the drain
- Gate current $i_g = 0$, drain current $i_d = g_m \times v_{gs}$, source current $i_s = v_{gs}/(1/g_m) = g_m \times v_{gs}$ (same as i_d)
- r_o can be added between drain and source if channel-length modulation must be included

KEY: T model is equivalent to pi model — both give identical terminal equations. T model is preferred when a resistance is connected in series with the source (CS with R_S , or CG amplifier), because the source current is obvious.

PART C: MOSFET AMPLIFIER CONFIGURATIONS

10. Three Basic MOSFET Amplifier Configurations

There are exactly three basic configurations for connecting the MOSFET as an amplifier. Each configuration is obtained by connecting one of the three MOSFET terminals to GROUND, making it common to both input and output ports.

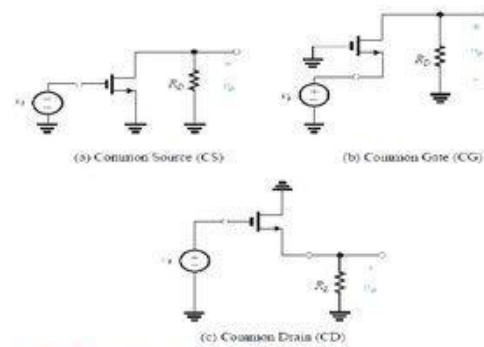


Figure 5.43 The three basic MOSFET amplifier configurations.

1. Common Source Amplifier (CS)

In the circuit of Fig. 5.43(a) the source terminal is connected to ground, the input voltage signal v_i is applied between the gate and ground, and the output voltage signal v_o is taken between the drain and ground, across the resistance R_D . This configuration, therefore, is called the grounded source or **common-source (CS)** amplifier. It is by far the most popular MOS amplifier.

2. Common Gate Amplifier (CG)

The **common-gate (CG)** or grounded-gate amplifier is shown in Fig. 5.43(b). It is obtained by connecting the gate to ground, applying the input v_i between the source and ground, and taking the output v_o across the resistance R_D connected between the drain and ground.

Figure 5.43: Three basic MOSFET amplifier configurations. (a) Common Source (CS) — source grounded. (b) Common Gate (CG) — gate grounded. (c) Common Drain (CD) / Source Follower — drain grounded.

Configuration 1 — Common Source (CS) Amplifier

- Source terminal is connected to ground (common terminal)
- Input applied between gate and ground
- Output taken between drain and ground across R_D
- Most popular MOSFET amplifier configuration — high voltage gain, infinite input resistance
- Produces 180° phase inversion between input and output

Configuration 2 — Common Gate (CG) Amplifier

- Gate terminal is connected to ground (common terminal)
- Input applied between source and ground
- Output taken between drain and ground across R_D
- Low input resistance ($= 1/g_m$), high output resistance — used for current buffering and impedance matching
- No phase inversion — positive voltage gain

Configuration 3 — Common Drain (CD) Amplifier / Source Follower

- Drain terminal is connected to ground (common terminal)
- Input applied between gate and ground
- Output taken between source and ground across load R_L

- Voltage gain ≈ 1 (unity) — acts as a voltage buffer
- Infinite input resistance, low output resistance ($= 1/g_m$)
- No phase inversion — called "source follower" because output (source) follows the input (gate)

11. Characterizing Amplifiers

11.1 Rin, Avo, Ro, Av, Gv Definitions

An amplifier is completely characterized by three parameters: input resistance, open-circuit voltage gain, and output resistance. Figure 5.44 shows the general amplifier model.

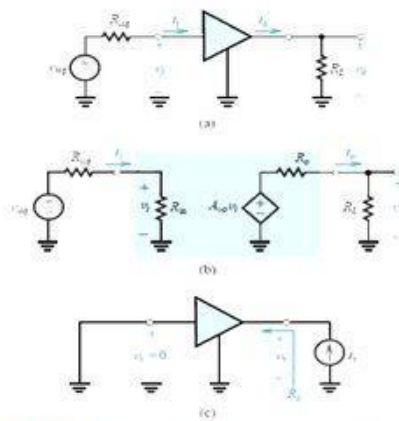


Figure 5.44 Characterization of the amplifier as a functional block. (a) An amplifier fed with a voltage signal v_{sig} having a source resistance R_{sig} and feeding a load resistance R_L . (b) Equivalent circuit representation of the circuit in (a). (c) Determining the amplifier output resistance R_o .

and together with the resistance R_{sig} forms a voltage divider that reduces v_{sig} to the value v_i that appears at the amplifier input.

$$v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig} \quad (5.45)$$

The second parameter in characterizing amplifier performance is the **open-circuit voltage gain** A_{vo} , defined as

$$A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L = \infty}$$

The third and final parameter is the output resistance R_o . Observe from Fig. 5.44(b) that R_o is the resistance seen looking back into the amplifier output terminal with v_i set to zero.

Figure 5.44: Amplifier characterization. (a) Amplifier with source v_{sig} , R_{sig} , and load R_L . (b) Equivalent circuit with R_{in} , A_{vo} , R_o . (c) Finding R_o by setting $v_i=0$.

Five key amplifier parameters:

- $R_{in} = v_i / i_i$ — Input resistance (loading effect on source)
- $A_{vo} = v_o / v_i \mid R_L = \infty$ — Open-circuit voltage gain (with no load connected)
- $R_o = v_x / i_x \mid v_i = 0$ — Output resistance (seen from output terminal, input set to zero)
- $A_v = v_o / v_i$ — Voltage gain with load R_L connected: $A_v = A_{vo} \times R_L / (R_L + R_o)$
- $G_v = v_o / v_{sig}$ — Overall voltage gain from source to output: $G_v = (R_{in} / (R_{in} + R_{sig})) \times A_{vo} \times R_L / (R_L + R_o)$

Key Equations:

$$\begin{aligned} v_i &= [R_{in} / (R_{in} + R_{sig})] \times v_{sig} \quad (\text{voltage divider at input}) \\ v_o &= [R_L / (R_L + R_o)] \times A_{vo} \times v_i \quad (\text{voltage divider at output}) \\ G_v &= [R_{in} / (R_{in} + R_{sig})] \times A_{vo} \times [R_L / (R_L + R_o)] \end{aligned}$$

12. CS Amplifier WITHOUT Source Resistance

12.1 Rin, Avo, Ro, Av, Gv Derivations

The CS amplifier (Figure 5.45a) with the MOSFET replaced by its pi model (Fig 5.45b). The source is grounded, gate receives input v_i , drain gives output v_o across R_D .

Figure 5.45(a) shows a common-source amplifier (with the biasing arrangement omitted) fed with a signal source v_{sig} having a source resistance R_{sig} . We wish to analyze this circuit to determine R_{in} , A_{vo} , R_o and G_v . For this purpose we shall assume that R_L is part of the amplifier; thus if a load resistance R_L is connected to the amplifier output, it appears in parallel with R_D .

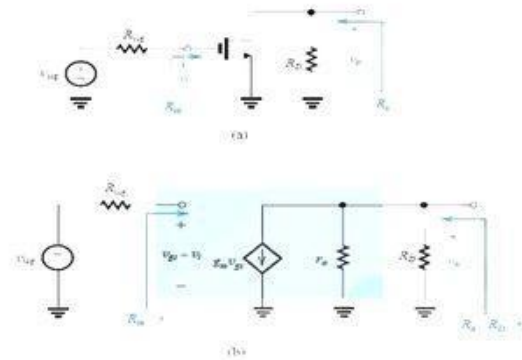


Figure 5.45 (a) Common-source amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig} . The bias circuit is omitted. (b) The common-source amplifier with the MOSFET replaced with its hybrid- π model.

Characteristic Parameters of the CS Amplifier Replacing the MOSFET with its hybrid- π model, we obtain the CS amplifier equivalent circuit shown in Fig 5.45(b). We shall use this equivalent circuit to determine the characteristic parameters R_{in} , A_{vo} , and R_o as follows.

The input resistance R_{in} is obviously infinite:

$$R_{in} = \infty \quad (5.69)$$

The output voltage v_o is found by multiplying the current $(g_m v_{gs})$ by the total resistance between the output node and ground.

Figure 5.45: (a) CS amplifier circuit with R_{sig} source. (b) AC equivalent using pi model — $R_{in} = \infty$, output resistance $R_o = R_D || r_o$.

Parameter 1 — Input Resistance Rin:

Since the MOSFET gate draws NO current, R_{in} is infinite:

$$R_{in} = \infty \quad (\text{MOSFET gate is an ideal open circuit})$$

Parameter 2 — Open-Circuit Voltage Gain Avo:

With $v_{gs} = v_i$ (since $R_{in} = \infty$, no loading at input), the output voltage is:

$$v_o = -(g_m \times v_{gs}) \times (R_D || r_o) = -g_m \times v_i \times (R_D || r_o)$$

$$A_{vo} = v_o/v_i = -g_m \times (R_D || r_o)$$

Approximation when $R_D \ll r_o$ (typical for discrete circuits):

$$A_{vo} \approx -g_m \times R_D$$

Parameter 3 — Output Resistance Ro:

Setting $v_i = 0 \rightarrow v_{gs} = 0 \rightarrow g_m v_{gs} = 0$ (current source is open). Looking back into drain:

$$R_o = R_D || r_o$$

With $R_D \ll r_o$: $R_o \approx R_D$

Parameter 4 — Overall Voltage Gain Gv:

Since $R_{in} = \infty$, $v_i = v_{sig}$ (no input signal loss). With load R_L :

$$A_v = -g_m \times (R_D || R_L || r_o) \approx -g_m \times (R_D || R_L)$$

$$G_v = v_o/v_{sig} = A_v = -g_m \times (R_D \parallel R_L)$$

NOTE: For the CS amplifier without R_S , the overall gain G_v equals A_v because the infinite input resistance causes no loading — the entire source voltage appears at the gate.

13. CS Amplifier WITH Source Resistance RS

13.1 Effect of RS and Gain Reduction

Inserting a resistance R_S in the source lead of the CS amplifier (Figure 5.47) is sometimes beneficial. R_S provides bias stability (DC), but it also degrades the AC gain. The T model is most convenient here because the source current flows through R_S explicitly.

Ans. ∞ ; -36.4 V/V; 18.2 k Ω ; -19 V/V; -19 V/V; 0.95 V

The Common-Source (CS) Amplifier with source resistance (R_S)

It is often beneficial to insert a resistance R_S in the source lead of the common-source amplifier as shown in Fig. 5.47(a). The corresponding small-signal equivalent circuit is shown in Fig. 5.47(b), where we note that the MOSFET has been replaced with its T equivalent-circuit model.

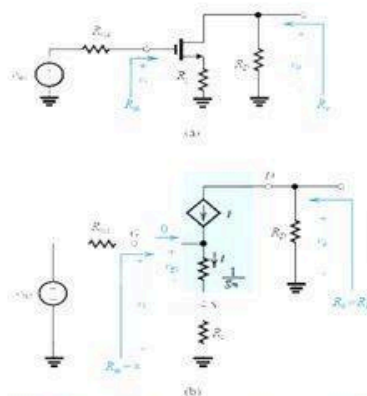


Figure 5.47 The CS amplifier with a source resistance R_S : (a) Circuit without bias details; (b) Equivalent circuit with the MOSFET represented by its T model.

Figure 5.47: (a) CS amplifier with source resistance R_S . (b) AC equivalent using T model — shows v_{gs} is reduced, not equal to v_i .

Since $R_{in} = \infty$, we have $v_i = v_{sig}$. However, the signal v_{gs} that controls the MOSFET is only a FRACTION of v_i due to the voltage divider formed by $1/g_m$ and R_S :

$$v_{gs} = v_i \times (1/g_m) / (1/g_m + R_S) = v_i / (1 + g_m R_S)$$

This shows that R_S reduces the effective gate-source signal voltage by the factor $(1 + g_m R_S)$.

13.2 A_{vo} , A_v , G_v for CS with R_S

Voltage Gain A_{vo} :

$$A_{vo} = v_o/v_i = -g_m R_D / (1 + g_m R_S)$$

Alternatively, expressing in terms of currents:

$$A_{vo} = -(R_D) / (1/g_m + R_S)$$

With load R_L :

$$\begin{aligned} A_v &= -g_m (R_D \parallel R_L) / (1 + g_m R_S) \\ &= -(R_D \parallel R_L) / (1/g_m + R_S) \end{aligned}$$

Output Resistance R_o :

$$R_o = R_D \text{ (same as CS without } R_S, \text{ since looking into drain with } R_S \text{ present)}$$

Overall Gain G_v ($R_{in} = \infty$, so $v_i = v_{sig}$):

$$G_v = -g_m \times (R_D \parallel R_L) / (1 + g_m \times R_S)$$

Key Observation — Trade-off:

- R_S reduces voltage gain by factor $(1 + g_m \times R_S)$ — significant gain loss
- But R_S stabilizes the DC Q-point and extends the bandwidth
- R_S also reduces nonlinear distortion — improves linearity
- For AC gain preservation while keeping DC stability: bypass R_S with a large capacitor (short for AC, open for DC)

14. The Common-Gate (CG) Amplifier

14.1 Circuit and T-Model Analysis

In the CG amplifier (Figure 5.48a), the gate is grounded, input is applied to the source, and output is taken from the drain. Since R_{sig} is in series with the source terminal, the T model is the most convenient equivalent circuit.

which can also be expressed as

$$A_{v, \text{sig}} = -\frac{R_D R_{in}}{1 + g_m R_{in}} \quad (5.80)$$

The output resistance R_o which can be found by inspection to be

$$R_o = R_D$$

Alternatively, $A_{v, \text{sig}}$ can be obtained by simply replacing R_D in Eq. (5.79) or (5.80) by $(R_D \parallel R_L)$; thus,

$$A_v = -\frac{R_D \parallel R_L}{1 + g_m R_{in}} \quad (5.82)$$

or

$$A_v = -\frac{g_m (R_D \parallel R_L) R_{in}}{1 + g_m R_{in}} \quad (5.83)$$

The Common-Gate (CG) Amplifier

Figure 5.48(a) shows a common-gate amplifier with the biasing circuit omitted. The amplifier is fed with a signal source characterized by v_i and R_{sig} . Since R_{sig} appears in series with the source, it is more convenient to represent the transistor with the T model than with the π model. Doing this, we obtain the amplifier equivalent circuit shown in Fig. 5.48(b). Note that we have not included r_o . This would have complicated the analysis considerably.

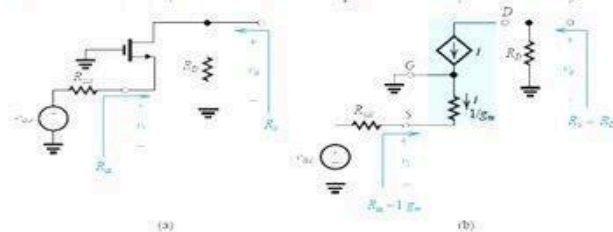


Figure 5.48 (a) Common gate (CG) amplifier with bias arrangement omitted. (b) Equivalent circuit of the CG amplifier with the MOSFET replaced with its T model.

Figure 5.48: (a) CG amplifier with gate grounded, input at source. (b) T-model equivalent — R_{sig} in series with $1/g_m$, $R_{in} = 1/g_m$.

From the T-model equivalent circuit:

- Gate is at AC ground. R_{sig} is in series with the source terminal.
- The $1/g_m$ resistance appears in series with R_{sig} to form the source loop.
- The controlled current source pumps current i into drain.

Input Resistance R_{in} :

Looking into the source terminal (with gate grounded):

$$R_{in} = 1/g_m$$

This is the key difference: CG has LOW input resistance ($1/g_m$ typically 100–500 Ω), unlike CS which has infinite R_{in} .

14.2 R_{in} , A_{vo} , R_o , A_v , G_v for CG

Open-Circuit Voltage Gain A_{vo} :

Writing at the drain node (with no load):

$$v_o = -i \times R_D \quad \text{where } i = -v_i / (1/g_m) = -g_m \times v_i$$

$$A_{vo} = v_o/v_i = g_m \times R_D$$

Note: POSITIVE gain — CG amplifier does NOT invert the signal!

Output Resistance R_o :

$$R_o = R_D$$

Voltage Gain A_v with load R_L :

$$A_v = g_m \times (R_D \parallel R_L)$$

Overall Voltage Gain G_v (input loading is significant!):

The low R_{in} of CG causes significant input signal attenuation. The input voltage divider gives:

$$\begin{aligned} v_i/v_{sig} &= R_{in}/(R_{in} + R_{sig}) = (1/g_m)/(1/g_m + R_{sig}) \\ G_v &= v_o/v_{sig} = (1/g_m \times g_m \times (R_D \parallel R_L)) / (1/g_m + R_{sig}) \\ G_v &= (R_D \parallel R_L) / (R_{sig} + 1/g_m) \end{aligned}$$

NOTE: The CG amplifier has low input resistance ($1/g_m$) which causes input signal to be attenuated when driven from a source with R_{sig} . However, CG has excellent high-frequency performance and is used in cascode amplifiers and RF circuits.

15. The Common-Drain (CD) / Source Follower

15.1 Circuit and T-Model Analysis

The Common-Drain amplifier (Figure 5.50) is also called the **source follower** because the output voltage at the source terminal "follows" the input at the gate. The drain is connected to VDD (AC ground). Input is at gate, output is at source across R_L .

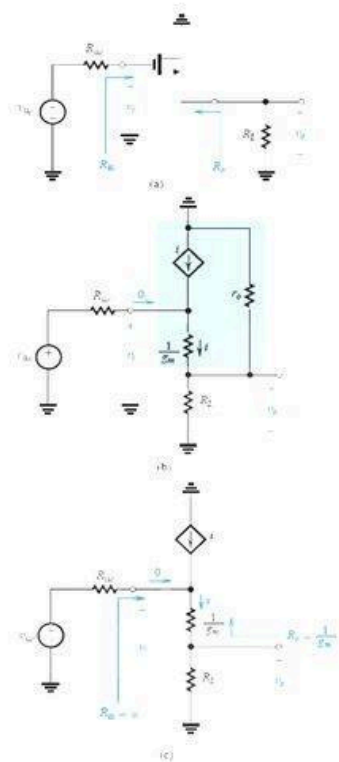


Figure 5.50 (a) Common-drain amplifier or source follower. (b) Equivalent circuit of the source follower obtained by replacing the MOSFET with its T model. Note that r_o appears in parallel with R_L and in this case circuit, $r_o = R_L$. Neglecting r_o , we obtain the simplified equivalent circuit in (c).

The unity open-circuit voltage gain together with R_L in Eq. (5.91) can be used to find A_i when a load resistance R_L is connected. The result is simply the expression in Eq. (5.89). Finally, because of the infinite R_{in} , $v_{gs} = v_i$, and the overall voltage gain is

$$G_v = A_i = \frac{R_L}{R_L + 1/g_m} \quad (5.92)$$

Figure 5.50: (a) Source follower circuit. (b) Equivalent using T model with r_o included. (c) Simplified equivalent — $r_o \gg R_L$ so r_o is neglected.

Using the T model (most convenient for CD), with r_o included in parallel with R_L :

- $R_{in} = \infty$ (gate draws no current — same as all MOSFET configurations)
- $v_i = v_{gs} + v_o$ (KVL around gate-source-output loop)
- Controlled current source i flows through the parallel combination of $1/g_m$, r_o , and R_L

15.2 A_v , R_o , G_v for Source Follower

Voltage Gain A_v :

From the circuit: $v_i = i(1/g_m + R_L)$ and $v_o = iR_L$ (voltage across R_L). Therefore:

$$A_v = v_o/v_i = R_L / (R_L + 1/g_m)$$

Setting $R_L = \infty$ (open circuit):

$$A_{vo} = 1 \quad (\text{unity open-circuit voltage gain})$$

Since $R_L \gg 1/g_m$ in most practical cases:

$$A_v \approx 1 \quad (\text{voltage gain is slightly less than 1})$$

Output Resistance R_o :

Setting $v_i = 0$ (grounding the gate) and looking into the source terminal:

$$R_o = 1/g_m$$

This very LOW output resistance makes the source follower excellent for driving low-impedance loads.

Overall Voltage Gain G_v :

Since $R_{in} = \infty$, $v_i = v_{sig}$ (no input loading). Therefore:

$$G_v = A_v = R_L / (R_L + 1/g_m)$$

Summary of Source Follower Properties:

- Voltage gain ≈ 1 (slightly less) — no voltage amplification, acts as buffer
- NO phase inversion — output follows input
- Input resistance = ∞ — ideal for receiving signal from any source
- Output resistance = $1/g_m$ — very low, ideal for driving low-impedance loads
- Current gain = very large (β of MOSFET is effectively infinite in this context)
- Primary application: Impedance transformation — between high-impedance source and low-impedance load

16. Comparison Table – All MOSFET Amplifier Configurations

Table 5.4 Characteristics of MOSFET Amplifiers					
Amplifier type	Characteristics ^a				
	R_{in}	$A_{v, no RS}$	R_{out}	$A_{v, RS}$	G_v
Common source (Fig. 5.15)	∞	$-g_m R_{D1}$	R_{D1}	$-g_m(R_{D1} \parallel R_{F1})$	$-g_m(R_{D1} \parallel R_{F1})$
Common source with R_{S1} (Fig. 5.17)	∞	$-\frac{g_m R_{D1}}{1 + g_m R_{S1}}$	R_{D1}	$-\frac{g_m(R_{D1} \parallel R_{F1})}{1 + g_m R_{S1}}$	$-\frac{g_m(R_{D1} \parallel R_{F1})}{1 + g_m R_{S1}}$
Common gate (Fig. 5.18)	$\frac{1}{g_m}$	$g_m R_{D1}$	R_{D1}	$g_m(R_{D1} \parallel R_{F1})$	$\frac{R_{D1} \parallel R_{F1}}{R_{S1} + 1/g_m}$
Source follower (Fig. 5.20)	∞	1	$\frac{1}{g_m}$	$\frac{R_{S1}}{R_{S1} + 1/g_m}$	$\frac{R_{S1}}{R_{S1} + 1/g_m}$

^a For the interpretation of R_{in} , $A_{v, no RS}$, and R_{out} , refer to Fig. 5.18(b).

^b The MOSFET output resistance r_{ds} has been neglected, as is permitted in the discrete circuit amplifiers studied in this chapter. For IC amplifiers, it must always be taken into account.

Table 5.4: Summary of MOSFET amplifier characteristics for CS, CS+RS, CG, and Source Follower.

Parameter	CS (no RS)	CS with RS	Common Gate (CG)	Source Follower (CD)
Rin	Infinity	Infinity	1/gm (LOW)	Infinity
Avo	-gm×RD	-gm×RD/(1+gm×RS)	gm×RD (positive)	1 (unity)
Ro	RD ro ≈ RD	RD	RD	1/gm (LOW)
Av (with RL)	-gm×(RD RL)	-gm(RD RL)/(1+gmRS)	gm×(RD RL)	RL/(RL+1/gm)
Gv	= Av (Rin=∞)	= Av (Rin=∞)	(RD RL)/(Rsig+1/gm)	= Av (Rin=∞)
Phase shift	180° (inverted)	180° (inverted)	0° (non-inverting)	0° (non-inverting)
Best use case	Voltage amplification	Gain+stability tradeoff	RF, cascode, current sense	Impedance matching, buffer

Quick Comparison Rules:

- CS: Highest voltage gain, 180° inversion, infinite Rin — most popular configuration
- CS+RS: Reduced gain but better stability and linearity — add bypass cap to restore gain for AC
- CG: Low Rin (=1/gm), no inversion, good at high freq — used in cascode circuits with CS
- CD (Source Follower): Unity gain, no inversion, infinite Rin, low Ro — ideal voltage buffer

17. Quick Revision – All Formulas & Exam Questions

Complete exam-focused summary for BEC303 Module 2 — MOSFET.

17.1 All Key Formulas

Formula	Description	Where Used
$I_D = (1/2) \times k_n \times (V_{GS} - V_t)^2 = (1/2) \times k_n \times V_{OV}^2$	DC drain current (saturation)	All biasing circuits
$V_{DS} = V_{DD} - R_D \times I_D$	DC drain-source voltage	All circuits
$V_{DS} > V_{OV}$	Saturation condition	Must verify Q-point is valid
$V_G = V_{GS} + R_S \times I_D$	Fixing-VG bias equation	VG bias and DTG feedback
$V_{GS} = V_{DS} = V_{DD} - R_D \times I_D$	Drain-to-gate feedback	Drain-to-gate biasing
$v_{GS} = V_{GS} + v_{gs}; v_{gs} \ll 2 \times V_{OV}$	Small-signal condition	Valid linear operation
$i_d = g_m \times v_{gs}$	Signal drain current	All small-signal analysis
$A_v = v_{ds}/v_{gs} = -g_m \times R_D$	CS voltage gain (basic)	CS small-signal analysis
$g_m = k_n \times V_{OV} = 2 \times I_D / V_{OV} = \sqrt{2 \times k_n' \times W/L \times I_D}$	Three gm expressions	Transconductance calculation
$r_o = V_A / I_D$	Output resistance (channel modulation)	Complete small-signal model
CS: $R_{in} = \infty, A_{vo} = -g_m \times R_D, R_o = R_D$	CS parameters	CS amplifier analysis
CS+RS: $A_{vo} = -g_m \times R_D / (1 + g_m \times R_S)$	CS with RS gain	CS+RS analysis
CG: $R_{in} = 1/g_m, A_{vo} = g_m \times R_D, R_o = R_D$	CG parameters	CG amplifier analysis
CG: $G_v = (R_D R_L) / (R_{sig} + 1/g_m)$	CG overall gain	CG with source loading
CD: $R_{in} = \infty, A_{vo} = 1, R_o = 1/g_m$	Source follower parameters	CD amplifier analysis
CD: $A_v = R_L / (R_L + 1/g_m); G_v = A_v$	Source follower gain	CD with load
$G_v = [R_{in} / (R_{in} + R_{sig})] \times A_{vo} \times [R_L / (R_L + R_o)]$	General overall gain	Any amplifier

17.2 Step-by-Step DC Analysis for Any MOSFET Bias Circuit

- Find V_{GS} (from biasing: $V_{GS} = V_G - V_S$, or $V_{GS} = V_{DS}$ for DTG)
- Calculate $I_D = (1/2) \times k_n \times (V_{GS} - V_t)^2$
- Calculate $V_{DS} = V_{DD} - I_D \times R_D$
- Verify $V_{DS} > V_{OV}$ (saturation check)
- Calculate $g_m = 2 \times I_D / V_{OV}$ or $g_m = k_n \times V_{OV}$
- Draw AC equivalent (short caps + DC sources, replace MOSFET with model)
- Find $R_{in}, A_{vo}, R_o, A_v, G_v$ from the AC equivalent

17.3 Top 12 Exam Questions

- Explain biasing by fixing V_{GS} . Why is it NOT a good approach? (6M)
- Explain biasing by fixing V_G with source resistance R_S . Explain the negative feedback mechanism. (8M)
- Explain drain-to-gate feedback biasing with circuit diagram and feedback mechanism. (6M)
- What is the DC bias point of a MOSFET amplifier? Derive I_D and V_{DS} equations. (4M)

26. Explain signal current in drain terminal. What is the small-signal condition? How is nonlinear distortion reduced? (6M)
27. Derive voltage gain of a MOSFET small-signal amplifier. Why is there a 180° phase inversion? (6M)
28. Draw and explain the small-signal pi equivalent circuit model of MOSFET. What is r_o ? (6M)
29. Define transconductance g_m . Write all three expressions for g_m and explain what each shows. (8M)
30. Explain the T equivalent circuit model. How is it derived from the pi model? When is T model preferred? (8M)
31. Explain the three basic MOSFET amplifier configurations with circuit diagrams. (6M)
32. Derive R_{in} , A_{vo} , R_o , A_v , G_v for CS amplifier without R_S using pi model. (8M)
33. Compare CS, CS+ R_S , CG, and source follower: R_{in} , A_{vo} , R_o , phase, application. Draw a table. (8M)

END OF MODULE 2 (BEC303 – MOSFET) – ALL THE BEST! 🎓

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